DIANA

Development of a susta**I**nable **A**morphous **N**ovel scintill**A**tor

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INFN
Istituto Nazionale di Fisica Nucleare



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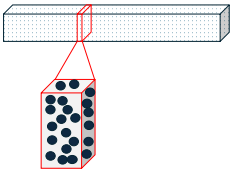
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Overview

- The scintillating materials find a wide use in many physics field
- Roughly, they can be divided into two categories: organic and inorganic scintillators
- The DIANA project aims to develop innovative scintillating materials combining the benefits of both organic and inorganic scintillators by incorporating scintillating crystals (SC) fragments in an amorphous medium (glass or polymer)
- These material could be employed in various physics fields, such as High Energy Physics and veto systems



Organic scintillators

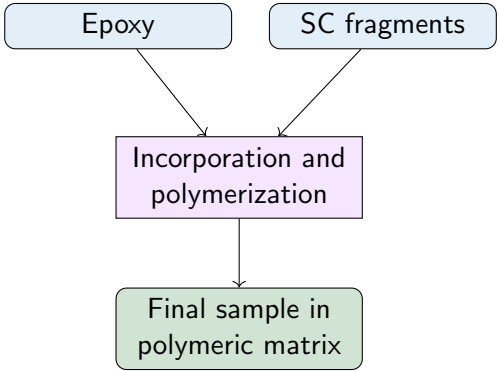
- ✓ Easily-tunable geometries
- ✓ Reduced production costs
- × Low density
- × Large radiation length

Inorganic scintillators

- ✓ High density
- ✓ Lower radiation length
- ✓ Lower Moliere radius
- × Complex production and high costs
- × Limited geometries of the final products

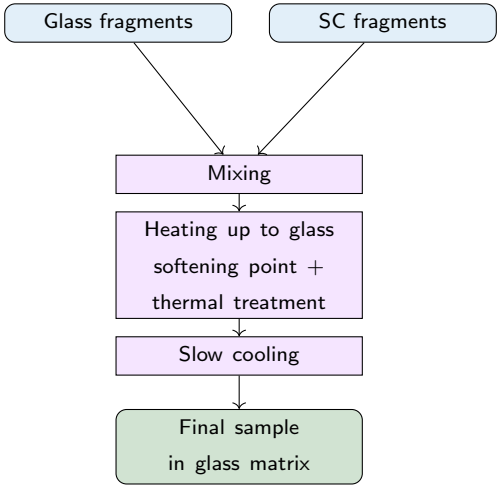
Materials and methodologies

- Bicomponent epoxy resin for initial studies → good transparency, the polymerization process doesn't require heat or high pressure
- Glass matrix:
  - Borosilicate glass (n $\simeq$ 1.5) → good compromise for the project
  - Lead glass (n up to 1.8) → high density, high refractive index



Materials and methodologies

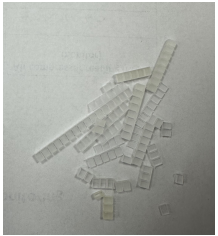
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Choice of the scintillating crystals

The SC must satisfy different requirements depending on the type of matrix:

- Polymeric matrix: SC with good light yield, with $\lambda_{\text{emission}}$ far from the absorption range of the matrix.
- Glass matrix: SC with good light yield, $\lambda_{\text{emission}}$ far from the absorption range, and softening point below the SC melting point.



Scintillator	T_m [°C]	n	ρ [g·cm^{-3}]	LY [γ·MeV^{-1}]	$\lambda_{emission}$ [nm]
$PbWO_4$	1123	2.17	8.28	170	420
CsI(Tl)	621	1.8	4.51	54000	550
NaI(Tl)	650	1.85	3.67	38000	415
LYSO(Ce)	1800	1.8	7.1	33000	420
LSO(Ce)	2150	1.8	7.41	33000	420
BGO	1050	2.15	7.1	8000	480

CsI and LYSO fragments will be provided by the Hilger Crystals company.

Construction of the prototypes

- Parallelepiped shape $2 \times 2 \times 20$ cm^3
- Possible insertion of optical fibers (plastic or quartz)
- Photons readout via SiPMs (Hamamatsu S13360)
- Reflective foils to maximize light gathering

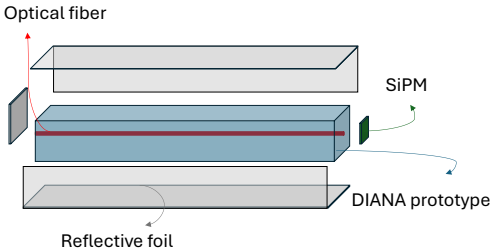


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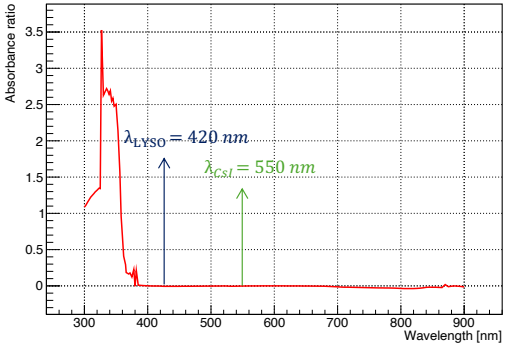
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Polymeric matrix

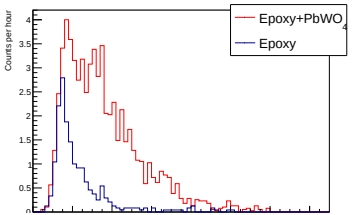
- Optical UV-Vis Absorbance test on a commercial epoxy resin
- Incorporation of $PbWO_4$ fragments in epoxy matrix. The result was a sample with medium density $\rho = 3.69 \pm 0.02\,\mathrm{g \cdot cm^{-3}}$ whose scintillation was tested using cosmic rays.

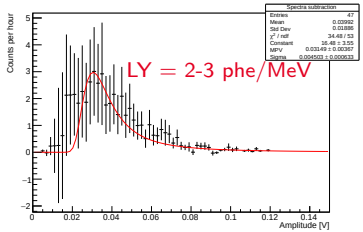


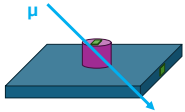
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- Incorporation of $PbWO_4$ fragments in epoxy matrix. The result was a sample with medium density $\rho = 3.69 \pm 0.02\,\mathrm{g \cdot cm^{-3}}$ whose scintillation was tested using cosmic rays.
- The result is very encouraging and shows the feasibility of the DIANA project









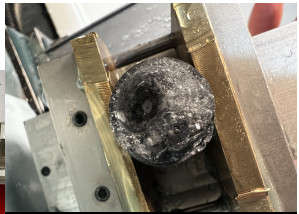
Glass matrix

- A $PbWO_4$ crystal was heated up to 800°C. After the test, this crystal showed unaltered scintillating properties
- A sample was prepared using lead glass and a fraction equal to 30%$_w$ of $PbWO_4$ fragments.

  The thermal cycle included:
  - A 20 minutes heating ramp from room temperature to 800-850°C
  - A 30 minutes long temperature dwell
  - Slow cooling down to room temperature (2 hours)

  The sample presented optical defects (air bubbles), but shows a correct inclusion of the SC fragments.





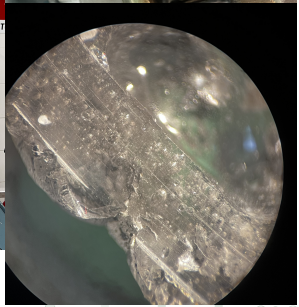


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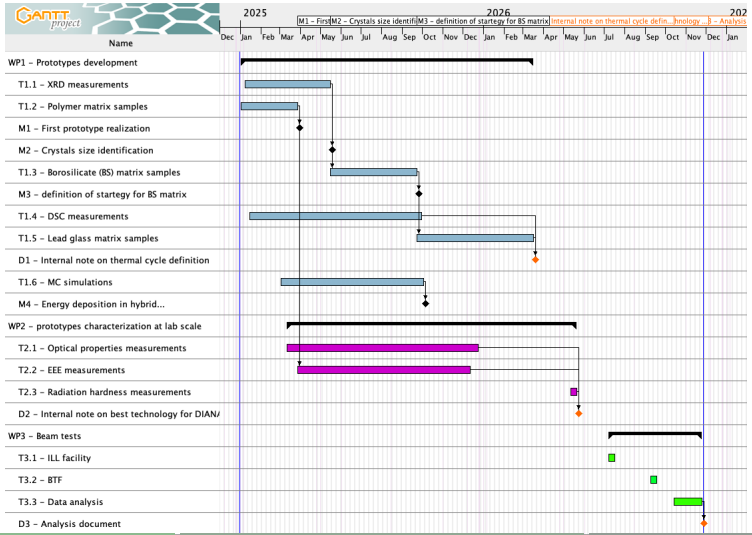
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Comprehensive schedule and milestones



The DIANA's team

Name	Institution	Role	FTE	Current position
A. Marini	INFN, Sezione di Genova	PI	0.9	Postdoc
M. Hamza	INFN, Sezione di Ferrara	XRD measurements	0.2	Ph.D. student
M. Romagnoni	INFN, Sezione di Ferrara	XRD measurements	0.1	RTDA
P. Manfrinetti	Università di Genova	chemical tests	0.1	Professor
M. Magnozzi	Università di Genova	optical measurements	0.1	RTDA
A. Celentano	INFN, Sezione di Genova	readout	0.05	INFN researcher
R. Spotorno	Università di Genova	chemical tests	0.05	Professor
Total			1.5	

Costs

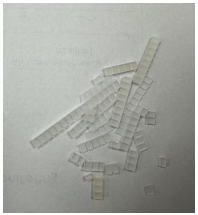
Voce	Costo unitario [€]	Quantità	Costo complessivo [€]
CsI(Tl) fragments	$0.07956 \cdot g^{-1}$	5 kg	397.80
LYSO(Ce) fragments	$0.57915 \cdot g^{-1}$	5 kg	2895.75
Nabertherm oven	10110	1	10110.00
SiPM S13360-6050PE	79,20	25	1980,00
SiPM S13360-3050PE	35,64	25	891.00
A5202 - 64 Channel Citiroc	5.473,71	1	5473,71
DAQ computer [a]	1000	1	1000,00
Missions	-	-	6000,00
Aquisition board [b]	6023,89	1	6023,89
Ferrara XRD maintenance	3000	1	3000
Total			≃40k

[a] Estimate based on the Test Point catalog, partnered with INFN.

[b] for the EEE telescope

Item:	Cost: $[€ \cdot g^{-1}]$
CsI(Tl) crystals	2.3
CsI(Tl) fragments	0.08
LYSO crystals	5.86
LYSO fragments	0.58

Quotation provided by Hilger Crystals. ▸ Link



Risk management

Risk sources	Consequences	P	I	Mitigation strategies
Delay in the delivery of materials for polymer matrix testing	Delay of the studies on polymer matrices	2	1	Use of the two-component epoxy resin available at INFN Genova
Delay in the delivery of SC	Delay of the studies on polymer matrices	2	1	Use of $PbWO_4$ crystals available at INFN Genova
Delay in the delivery of glass	Delay of the studies on glass matrices	2	1	Use of the lead glass available at INFN Genova
Delay in delivery of readout instrumentations	Delay of scintillation tests	2	1	Use of SiPMs and electronic modules already available at INFN Genova
Unavailability of the XRD from INFN Ferrara	Impossibility to perform the fragments size characterizations	2	3	Use of the Genova's Physics Department resources
Difficulties in the mixing strategy of SC fragments and glass	Inability to produce the material in glassy matrix	2	5	Use of samples in polymer matrix
Unavailability of the Gießen, BTF, and ILL facilities within the expected timelines	Delay in radiation resistance testing	2	3	Flexible timing has been planned for measurements at external facilities.

Color	Code	Incidence probability (P)	Impact on schedule (I)
	1	Low	Negligible
	2	Medium-low	$\simeq$ 1 month
	3	Medium	$\simeq$ 1-2 months
	4	High	$\simeq$ 2-4 months
	5	Very high	Supension of some goals

This shows the capability to adapt the DIANA project even in the worst-case scenarios

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Conclusions

- The DIANA project aims to the realization of novel scintillators based on scintillating crystals in an amorphous medium, reaching $3 - 4$ g·cm^{-3} densities
- The feasibility of DIANA is shown by the preliminary tests
- This novel material is easily workable and economically competitive
- This material can be employed in high energy physics experiments and as veto system (for example in rare-events physics).
- The synergies with the EEE collaboration and the Gießen University are already granted by supportive letters to the DIANA project
- The SC procurement at a competitive price is granted by the collaboration with Hilger Crystals.
- The risk analysis enlights the possibility to adapt the project even in the worst-case scenario.

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